National University of Computer and Emerging Sciences (FAST-NUCES)

PROJECT REPORT

ARTIFICIAL INTELLIGENCE

AI Smart-Distiller & Workflow Automator

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AI Smart-Distiller & Workflow Automator

1. Introduction

In today's digital era, students and professionals are constantly exposed to large volumes of information such as articles, lectures, reports, and project briefs. Extracting useful insights from such data is time-consuming and inefficient.

The AI Smart-Distiller & Workflow Automator is designed to solve this problem by transforming raw text into structured, actionable insights using Artificial Intelligence. The system not only summarizes content but also automates workflows by sending extracted data to productivity platforms.

2. Problem Statement

The major problem addressed in this project is Information Overload.

Users often face difficulties in:

- Identifying important tasks and deadlines
- Summarizing long documents
- Manually transferring information to tools like spreadsheets or communication apps

This leads to:

- Time wastage
- Human errors
- Reduced productivity

3. Objectives

The main objectives of this project are:

- To develop an AI-powered system for extracting key insights from text
- To automate task management and workflow execution
- To reduce manual effort in organizing information
- To improve productivity using intelligent automation

4. System Overview

The system acts as an Information Pipeline that processes input text and produces structured outputs.

Key Features:

- AI-based text summarization
- Extraction of tasks, deadlines, and key points
- Automated workflow execution
- Integration with external tools (Google Sheets, Discord, Email)

5. Methodology

The system follows a multi-step pipeline:

Step 1: Input

- Text (articles, notes, etc.)
- OR URL

Step 2: Processing

Text is sent to an AI model via API. Natural Language Understanding is applied.

Step 3: Data Structuring

- Summary
- Tasks
- Deadlines

Step 4: Automation

JSON is sent to automation tool (n8n). Data is forwarded to external platforms.

Step 5: Output

- Summary displayed on UI
- Tasks stored in Google Sheets
- Notifications sent via Discord/Email

6. Technologies Used

1. OpenAI API

Used for Natural Language Processing. Extracts structured insights from text.

2. Streamlit

Used to build the frontend interface. Provides user interaction.

3. n8n

Handles automation workflows. Connects with external services.

4. Python Libraries

- Requests (API communication)
- JSON (data formatting)

7. System Architecture

The system consists of three main components:

- Frontend (Streamlit): Accepts user input and displays output
- AI Processing Layer: Uses OpenAI API to convert text into structured data
- Automation Layer (n8n): Executes workflows and integrates with third-party apps

8. Innovations

LLM-Based Understanding: Unlike keyword-based systems, this project uses AI to understand context and meaning.

End-to-End Automation: Automatically transfers extracted data to external tools.

Real-Time Processing: Instant transformation of text into actionable insights.

9. Results and Expected Outcomes

- Accurate summaries of long text
- Extraction of relevant tasks and deadlines
- Automated data entry into external systems
- Reduction in manual effort by up to 90%

10. Advantages

- Saves time and effort
- Reduces human error
- Improves productivity
- Easy-to-use interface

11. Limitations

- Depends on API availability

- Requires internet connection
- Accuracy depends on input quality

12. Future Enhancements

- Integration with more platforms (e.g., Notion, Trello)
- Voice input support
- Mobile application version
- Advanced analytics dashboard

13. Conclusion

The AI Smart-Distiller & Workflow Automator successfully demonstrates how Artificial Intelligence can be used to bridge the gap between information consumption and action. By combining AI with workflow automation, the system enhances productivity and reduces manual effort significantly.